Focusly

GenAI Animated Learning Platform

Phase 1

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Phase 1

Implementation-focused overview

What is Focusly?

Focusly converts a topic prompt into a short, visually animated lesson video. The phase-1 goal is a working GenAI pipeline that plans the lesson, writes the script, generates scenes, adds narration, and renders the final video.

Prompt	**User enters one topic**
AI Plan	**Script, scenes, quiz checkpoints**
Video	**Animated lesson output**

Implementation Stack

The tools are selected to make a working pipeline possible without overbuilding.

Layer	Tool	Phase 1 Role	Status
Frontend	Next.js + React	Prompt form, dashboard, lesson player	MVP
Backend	FastAPI	Auth, lesson APIs, job polling	MVP
Queue	Redis + ARQ	Async generation jobs	MVP
Database	PostgreSQL	Users, jobs, metadata, checkpoints	MVP
Storage	Cloudflare R2	Generated video and assets	MVP
Observability	Sentry + logs	Errors, cost, generation status	Basic

GenAI Model Choice

High-level model usage for the evaluation, without going deep into model internals.

Primary LLM

Claude API via LangChain

Used for script, structured outputs, scene planning, and code generation.

Why this choice

LangChain gives prompt templates, Pydantic parsing, retry handling, tools, and cost tracking callbacks.

Inference only

No model training in phase 1

Minimum Viable Product Scope

input, generation, narration, video output.

✓ Topic Input

Student enters a concept such as binary search, sorting, or DBMS normalization.

✓ AI Lesson Planning

LLM creates objectives, pacing, script, scene plan, and quiz checkpoints.

✓ Video Rendering

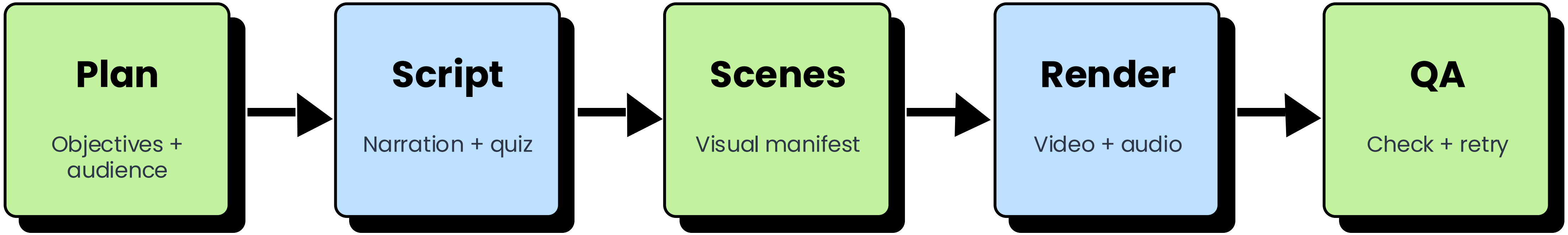
Remotion and Manim generate controlled educational animation scenes.

✓ Delivery

Final MP4/HLS stream is stored and shown in the Focusly player.

AI Pipeline Overview

LangGraph controls the workflow; LangChain wraps the model calls and tools.



Video Generation Tools

Output is generated through controllable code based animation, not black-box AI video generation.

Remotion

React-based animated scenes, text reveals, quiz screens, and general visual layouts.

Manim

Math, algorithms, graphs, and step-by-step concept animation.

FFmpeg

Stitches scenes, mixes narration, encodes MP4, and prepares HLS streaming.

ElevenLabs — TTS narration

Cloudflare R2 — Video storage

Phase 1 Evaluation Focus

implementation choices, tools, model usage, and demo scope.

Area	What We use	Depth	Priority
Model	Claude API via LangChain	High level	High
Pipeline	LangGraph flow from prompt to render	High level	High
Video	Remotion, Manim, FFmpeg, ElevenLabs	Tool level	High
Backend	FastAPI, Redis queue, PostgreSQL	Tool level	Medium
Out of scope	Deep model theory and post-MVP features	Avoid	Clear